where each α_j is a non-negative integer. The vector α is called a multi-index of order

$$|\alpha| := \alpha_1 + \cdots + \alpha_d.$$

Then we define

$$D^\alpha u(x) := \frac{\partial^{|\alpha|} u(x)}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}.$$

In particular, we define the set of derivatives of order k as

$$D^k u(x) := \{D^\alpha u(x) : \ |\alpha| = k\}$$

If we assign a particular ordering to such a set, we may define a tuple $(D^{\alpha_1} u(x), \dots, D^{\alpha_{n_k}} u(x))$, where n_k depends on the order k of differentiation, and interpret the above set of all partial derivatives of order k as a point in $\mathbb{R}^{n_k}$. This notation will be convenient for writing PDE in a general way.

The special case of mappings between real (or complex) numbers (or, more generally, vectors thereof) are so common, we shall reserve the term "function" for this case. For mappings between general X and Y, we stick with the generic term, and say f defines a "mapping" or say that f is an "operator".

Yet another special case of nomenclature is usually reserved for the case that a function is mapped to the real (or complex) numbers. Such mappings $l : V \to \mathbb{R}$ are often called functionals. Note that this is, in general, different from a simple function between the real numbers, as the set V can be more general.

A.2.2 Order Symbols

We often need to estimate the magnitude of a function by comparing it to another function. Order symbols are merely a convenient notation for this. We give a formal definition for functions defined on the real numbers. There are rather obvious extensions, for example for sequences. Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions, and let $x_0 \in \mathbb{R}$. We say

$$f(x) = \mathcal{O}(g(x)) \qquad (\text{as } x \to x_0), \tag{A.5}$$

(read: "f is of order g, as x goes to x_0") if there are positive constants C, δ, such that

$$|f(x)| \leq C|g(x)|, \qquad \text{whenever } |x - x_0| < \delta, \tag{A.6}$$

We say that f is of order g as x goes to infinity, if there exists a positive constant C and $\overline{x} \in \mathbb{R}$, such that

$$|f(x)| \leq C|g(x)|, \qquad \text{for } x > \overline{x}. \tag{A.7}$$

Note that certain limits (e.g. $x \to x_0$) are involved in the preceding definitions. Often it is clear from the context which limit is to be taken, and we simply say "f is of order g", without stating or writing the limit explicitly. Also note that, in the statement $f(x) = \mathcal{O}(g(x))$ in (A.5), the constant C in the definition (A.6) or (A.7) is not included. This is because usually the function g by which we can bound the function f is of far greater importance than the constant involved.

A.3 Complex Numbers

The extension of real numbers to complex numbers begins with the *imaginary unit*

$$\hat{\imath} := \sqrt{-1}.$$

Historically, complex numbers arose in the context of solving algebraic equations. For example, no real number is the root of $x^2 + 1 = 0$, but using the imaginary unit defined above, we have two roots, given by $x = \pm\hat{\imath}$. We may extend the imaginary unit to imaginary numbers by

multiplication with any $y \in \mathbb{R}$. We call $\hat{\imath}y$ an imaginary number. The combination of a real number and an imaginary number is a complex number,

$$z := x + \hat{\imath}y, \qquad x, y \in \mathbb{R},$$

where $Re(z) = x$ is the real part, and $Im(z) = y$ is the imaginary part. (Note carefully: $\hat{\imath}y$ is an imaginary *number*, which is distinguished from the imaginary *part* y, which is a real number.)

Adding, subtracting and multiplying complex numbers follows "standard" arithmetic[2]. For example, we can write for the addition of two complex numbers

$$z_1 + z_2 = (x_1 + \hat{\imath}y_1) + (x_2 + \hat{\imath}y_2) = (x_1 + x_2) + \hat{\imath}(y_1 + y_2).$$

It is easy to see that complex numbers using "standard" arithmetic form a field of numbers, just like the real numbers do. We usually denote the field of complex numbers as $\mathbb{C} := \{z : z = x + \hat{\imath}y; \hat{\imath} = \sqrt{-1}; x, y \in \mathbb{R}\}$.

We define the complex conjugate of $z = x + \hat{\imath}y$ as

$$\overline{z} := x - \hat{\imath}y.$$

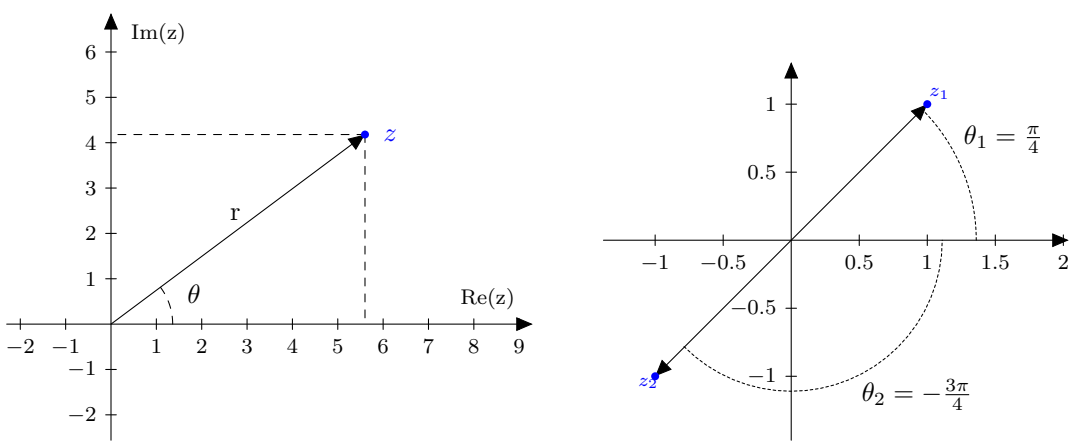
The complex conjugate allows us to extract the real and imaginary parts of a complex number z by using algebraic operations:

$$Re(z) = \frac{1}{2}(z + \overline{z}),$$
$$Im(z) = \frac{1}{2\hat{\imath}}(z - \overline{z}).$$

We define the norm, or magnitude, of a complex number as

$$|z| := \sqrt{\overline{z}z} = \sqrt{x^2 + y^2}.$$

This is consistent with the interpretation of a complex number as a point in the Cartesian



(a) A complex number in the Cartesian plane

(b) Two complex numbers with $Re(z)/Im(z) = 1$, but different argument.

Figure A.1: Illustration of complex numbers.

plane $\{(x, y) : x, y \in \mathbb{R}\}$ where $x = Re(z)$ and $y = Im(z)$. Figure A.1(a) shows an illustration. This characterization is often helpful in understanding the behavior of complex numbers. It also motivates a different representation of complex numbers: Instead of defining a complex number by the real part and the imaginary part, we may give the magnitude and an angle with respect

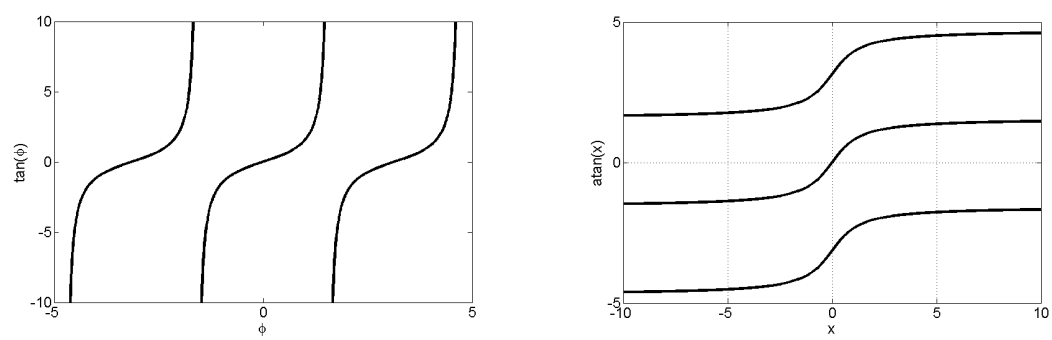
[2]By standard arithmetic we mean that imaginary unit is treated just like a real number in arithmetic operation, except we have to remember to set $\hat{\imath}^2 = -1$

to a reference direction in the Cartesian plane. This is also shown in Fig. A.1(a) for the usual case of taking the reference direction as the positive real axis. We then use $r := \sqrt{x^2 + y^2}$ and θ such that $\tan\theta = y/x$. We call θ the argument of z.

Remark A.1 Note that getting the the correct value of the argument can be a little tricky. If you take $z_1 = 1 + \hat{\imath}$ and $z_2 = -1 - \hat{\imath}$, both have $y/x = 1$, due to cancellation of the minus sign. If we define the argument by simply using the inverse tangent, we would get $\arg(z_1) = \arg(z_2) = \pi/4$: The standard definition of the inverse tangent always takes values on the "principal branch". (See the illustration in Fig. A.2.) The key to getting the correct argument is to not always take the principal value of the inverse tangent, which is always between $-\pi/2$ and $\pi/2$, but to take the argument from the correct branch, based on the quadrant in which your complex number is located. In a variety of computer languages this is implemented as `atan2`:

$$\begin{aligned} \text{atan2} : \mathbb{R}^2 \backslash \{(0,0)\} &\to (-\pi, \pi] \\ (x,y) &\mapsto \begin{cases} \tan^{-1}\left(\frac{y}{x}\right) & x > 0 \\ \tan^{-1}\left(\frac{y}{x}\right) + \pi & x < 0, y \geq 0 \\ \tan^{-1}\left(\frac{y}{x}\right) - \pi & x < 0, y < 0 \\ \frac{\pi}{2} & x = 0, y > 0 \\ -\frac{\pi}{2} & x = 0, y < 0 \end{cases} \end{aligned}$$

Note that $\text{atan2}(0,0)$ is undefined. With this definition, we get $\arg(z_1) = \pi/4$ and $\arg(z_2) = -3\pi/4$.



(a) The tangent function

(b) Branches of the inverse tangent function

Figure A.2: Graphs for the tangent and the inverse tangent. Note that there are several "branches" of the inverse tangent.

Given the characterization using magnitude and argument, the real part and the imaginary part can be written as

$$\begin{aligned} Re(z) &= x = r\cos\theta \\ Im(z) &= y = r\sin\theta. \end{aligned}$$

A useful way of writing a complex number is thus given by the polar form

$$z = re^{\hat{\imath}\theta}. \tag{A.8}$$

It can be readily seen from the series definition of the exponential, and that of the sine and cosine function that

$$e^{\hat{\imath}\theta} = \cos\theta + \hat{\imath}\sin\theta$$

So (A.8) is just another way of saying $z := Re(z) + \hat{\imath}Im(z)$. It is useful because the exponential function has many nice properties that allow easy manipulation of complex numbers when written in this form. For example, multiplication of complex numbers is easier using the polar form:

$$z_1 z_2 = r_1 e^{\hat{\imath}\theta_1} r_2 e^{\hat{\imath}\theta_2} = r_1 r_2 e^{\hat{\imath}(\theta_1+\theta_2)}$$

To multiply two complex numbers we thus have to multiply the magnitudes and add the arguments to obtain the result in polar form.

A.4 Review of Functional Analysis

Throughout this chapter V denotes a set, and $\mathbb{K}$ is either the field $\mathbb{R}$ or $\mathbb{C}$.

A.4.1 Linear Spaces

Definition A.1 *Assume that for any $u, v \in V$, an operation $(u, v) \mapsto u + v$ is defined, such that $u + v \in V$. Furthermore, assume that for any $v \in V$, and any $\alpha \in \mathbb{K}$, an operation $(\alpha, u) \mapsto \alpha u$ is defined, such that $\alpha u \in V$. We call these operations* addition, *and* scalar multiplication, *respectively. We call V a linear space, provided that for any $u, v, w \in V$, and $\alpha, \beta \in \mathbb{K}$, the following properties hold:*

- *Associativity of addition:* $u + (v + w) = (u + v) + w$
- *Commutativity of addition:* $u + v = v + u$
- *Identity element of addition:* *There exists an element $0 \in V$, such that $u + 0 = u$.*
- *Inverse element of addition:* *There exists an element $-u \in V$, such that $u + (-u) = 0$*
- *Distributivity of scalar multiplication with respect to vector addition:* $\alpha(u+v) = \alpha u + \alpha v$
- *Distributivity of scalar multiplication with respect to field addition:* $(\alpha+\beta)u = \alpha u + \beta v$
- *Compatibility of scalar multiplication with field multiplication:* $\alpha(\beta u) = (\alpha\beta)u$
- *Identity element of scalar multiplication: There exists an element $1 \in \mathbb{K}$ such that $1v = v$*

Sometimes the term *vector space* is used instead of linear space. We prefer to use the term linear space, however, since it is more descriptive: The set V is closed with respect to the linear combination $\alpha u + \beta v$. Furthermore, we should carefully avoid confusion with the Euclidean vector space, which is merely one example of the general definition A.1:

Example A.1 It is easy to verify that the following sets fulfill the properties of Definition A.1, if the respective "standard" definitions of addition and scalar multiplication are used:

- $V = \mathbb{K}^n$, where $n \in \mathbb{N}$. (Euclidean vector space.) For $\mathbf{x}, \mathbf{y} \in \mathbb{K}^n$

$$\mathbf{x} + \mathbf{y} := \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}, \qquad \alpha\mathbf{x} := \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix}.$$

Note that, in particular, the field $\mathbb{K}$ itself is a linear space.

- $V = P^m(I)$. (Polynomials of degree m defined on some $I \subset \mathbb{R}$.) Addition and scalar multiplication are defined in the usual "pointwise" sense, i.e. for $p, q \in V$,

$$(p+q)(x) = p(x) + q(x), \qquad \forall x \in I, \tag{A.9}$$
$$(\alpha p)(x) = \alpha p(x), \qquad \forall x \in I. \tag{A.10}$$

It is easy to see that the sum of two polynomials is again a polynomial. Indeed, if we write $p(x) = \sum_{k=1}^m a_k x^k$ and $q(x) = \sum_{k=1}^m b_k x^k$, then

$$p(x) + q(x) = \sum_{k=1}^m \underbrace{(a_k + b_k)}_{c_k} x^k$$
$$\alpha p(x) = \sum_{k=1}^m \underbrace{(\alpha a_k)}_{d_k} x^k$$

- $V = C(I)$. (Continuous functions defined on some $I \subset \mathbb{R}$.) Addition and multiplication are again defined as in (A.9), (A.10).

Of course these definitions can be easily generalized (e.g. define polynomials or continuous functions on $\mathbb{K}^d$), and we shall encounter many more examples. Perhaps it is also instructive to give a counterexample:

Example A.2 The set $V = \{p \in P^m([0,1]) : p(0) = 1\}$ is *not* a linear space. (Evidently, for $p, q \in V$, the sum $p + q$ is not in V.)

Definition A.2 *Let V be a linear space. We say $v_1, v_2, \ldots, v_n \in V$ are* linearly independent *if, for $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{K}$,*

$$\sum_{k=1}^n \alpha_k v_k = 0 \quad \Longleftrightarrow \quad \alpha_1 = \alpha_2 = \cdots = \alpha_n = 0,$$

Otherwise we say that the $v_1, v_2, \ldots, v_n \in V$ are linearly dependent.

The only way to obtain the zero element of V with a linear combination of element that are linearly independent is the trivial one. Note that this implies that a collection of elements of V that contains the zero element, is never linearly independent. Also, any subset of a set of linearly independent elements is again linearly independent. Likewise, if we add any number of elements to a collection of linearly dependent elements, the larger set is still linearly dependent.

Example A.3 Let $f, g : \mathbb{R} \to \mathbb{R}$, and $a, b \in \mathbb{R}$. Consider

$$f(x) = ax$$
$$g(x) = bx.$$

These functions may be viewed, for example, as elements of $C(\mathbb{R})$ or of $P^m(\mathbb{R})$ for any m. These two functions are linearly dependent, as

$$bf(x) - ag(x) = 0, \qquad \forall x \in \mathbb{R}.$$

On the other hand, if we define two functions $f, g \in P^m(\mathbb{R})$ $(m > 2)$ as

$$\begin{aligned} f(x) &= x \\ g(x) &= x^2 \end{aligned}$$

then these functions are linearly independent. To see this one needs to show that the expression

$$\alpha f(x) + \beta g(x) = 0, \quad \forall x \in \mathbb{R}, \tag{A.11}$$

can only be satisfied for $\alpha = \beta = 0$. Assume that (A.11) holds, and differentiate twice to get

$$\alpha f''(x) + \beta g''(x) = 2b = 0 \quad \Longleftrightarrow \quad \beta = 0.$$

(Note that $f'' = 0$.) But then (A.11) imples that $\alpha = 0$.

Definition A.3 *A linear space V is* finite dimensional *if there is $n \in \mathbb{N}$, such that any collection $v_1, \ldots, v_m \in V$ is linearly dependent for $m > n$. If, in addition, there are linearly independent vectors $u_1, \ldots, u_n$ we say that V has dimnension n. The set $u_1, \ldots, u_n$ is called a basis (for V).*

Note that, if V is a finite-dimensional linear space, then any $u \in V$ can be written as a linear combination of the basis elements:

$$u = \sum_{k=1}^{n} c_k u_k, \qquad c_k \in \mathbb{K}. \tag{A.12}$$

Indeed, $u_1, \ldots, u_n, u$ must be linearly dependent for *any* $u \in V$, and consequently there must be some coefficients $a_k \in \mathbb{K}$ for $k = 1, \ldots, n+1$, not all zero, such that

$$\sum_{k=1}^{n} a_k u_k + a_{n+1} u = 0.$$

Now, the coefficient a_{n+1} cannot be zero, because otherwise we would get a non-trivial representation of the zero element by the basis elements, which is a contradiction. (The basis elements must be linearly independent.) So we can divide by a_{n+1}, and obtain (A.12) with $c_k = -a_k/a_{n+1}$.

Example A.4 Let us note a few examples of finite-dimensional spaces:

- $V = \mathbb{K}^n$ $(n \in \mathbb{N})$ is n-dimensional. The standard Cartesian unit vectors

  $$\mathbf{e}_i = (0, 0, \ldots, 0, 1, 0, \ldots, 0)^T$$

  (where the 1 is in the ith place) form a basis.

- $V = P^m(\mathbb{R})$ ($m \in \mathbb{N}$) is $m+1$-dimensional. The *monomials* $u_k = x^k$ form a basis. Using a similar approach to that used in Example A.3 one can show that these functions are linearly independent, and certainly any polynomial of degree m can be written

$$p(x) = \sum_{i=0}^{m} c_i x^i, \qquad c_i \in \mathbb{R}.$$

- $V = P(\mathbb{R})$, the space of *all* polynomials, $P(I) = \{p \in P^m(I) : m = 0, 1, 2, \dots\}$, is not finite-dimensional: The degree m can be arbitrarily large, and hence we cannot find a finite number of monomials (which are linearly independent) that can represent any polynomial.
- $V = C(I)$ ($I \subset \mathbb{K}$) is of infinite dimension. All continuous functions cannot be represented by any finite collection of basis functions.

A.4.2 Normed Spaces

Definition A.4 *Let V be a linear space. A* norm *is a mapping $\|\cdot\| : V \to R$ with the following properties: For all $u, v \in V$ and all $\alpha \in K$ there holds*

$$\text{Definiteness}: \quad \|v\| > 0 \text{ for } v \neq \mathbf{0}. \tag{A.13}$$

$$\text{positive Homogeneity}: \quad \|\alpha v\| = |\alpha| \, \|v\| \tag{A.14}$$

$$\text{Triangle Inequality}: \quad \|v + u\| \leq \|v\| + \|u\| \tag{A.15}$$

We call $(V, \|\cdot\|)$ a normed linear space. Frequently, when the choice of norm is clear from the context, we omit the symbol $\|\cdot\|$.

Example A.5

- $V = \mathbb{K}^n$. The mapping

$$\mathbb{K}^n \ni \mathbf{v} \mapsto \left(\sum_{i=1}^{n} |v_i|^2 \right)^{\frac{1}{2}} =: |\mathbf{v}| \tag{A.16}$$

satisfies the norm properties of Definition A.4. (We prove this below in a more general setting.) This is the standard Euclidean vector norm. Note that we use the modulus notation $|\cdot|$ to denote a norm in $\mathbb{K}^n$. (The standard modulus for $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$, which is also used in (A.16), is just a special case for $n = 1$.) One can generalize (A.16) to define the *p-norm*, for $1 \leq p < \infty$, as

$$|\mathbf{v}|_p := \left(\sum_{i=1}^{n} |v_i|^p \right)^{\frac{1}{p}}. \tag{A.17}$$

It may be shown that this definition satisfies all the properties (A.13) through (A.15). In the limit $p \to \infty$ one obtains $|\mathbf{v}|_\infty = \max_i |v_i|$. This latter norm is also sometimes called *maximum norm* for obvious reasons. Thus (A.16) is seen to be a special case of (A.17). However, we shall most often omit the subscript $p = 2$ when using the standard Euclidean norm (A.16).

- $V = C([0,1])$. The mapping

$$V \ni v \mapsto \|v\| \equiv \|v\|_\infty := \max_{0\le x\le 1} |v(x)| \tag{A.18}$$

satisfies the properties of Definition A.4. The only property which is slightly non-trivial to prove is the triangle inequality. For any $u, v \in V$ we have

$$\begin{aligned}\|u+v\| &= \max_{0\le x\le 1} |u+v| \\ &\le \max_{0\le x\le 1} \left| \max_{0\le x\le 1} |u| + \max_{0\le x\le 1} |v| \right| \\ &= \max_{0\le x\le 1} |u| + \max_{0\le x\le 1} |v| \\ &= \|u\| + \|v\| .\end{aligned}$$

Another choice of norm is given by

$$V \ni v \mapsto \|v\| := \int_0^1 |v|\, dx.$$

Again, homogeneity and definiteness follow directly from the definition. The triangle inequality can be proved using the fact that the modulus satisfies the triangle inequality: For any $v, w \in V$ we have

$$\|v+w\| = \int_0^1 |v+w|\, dx \le \int_0^1 |v| + |w|\, dx = \|v\| + \|w\| .$$

Given that there are apparently many different choices, we may ask if the choice of norm makes any difference. The short answer is "yes", but it does depend on the context. For finite-dimensional space one can show at least that all norms are *equivalent*, which means that there exist real constants $c, C > 0$, such that

$$c\,\|v\|_a \le \|v\|_b \le C\,\|v\|_a , \qquad \forall v \in V,$$

if both $(V, \|\cdot\|_a)$ and $(V, \|\cdot\|_b)$ are proper normed spaces. Generally speaking, however, we shall be concerned with infinite-dimensional spaces. (The solutions to PDE problems usually lie in infinite-dimensional spaces.) Here, not all norms are equivalent, and the "right" choice of norm usually has to do with *limits* of sequences. We start with a few definitions.

Definition A.5 *Let $(V, \|\cdot\|)$ be a normed space, and consider the sequence $\{u_n\}_{n\in\mathbb{N}} \subset V$. We say the sequence converges in V if there is $u \in V$ such that*

$$\lim_{n\to\infty} \|u_n - u\| = 0,$$

which is to say that, for any $\varepsilon > 0$, there is $N_\varepsilon \in \mathbb{N}$, such that

$$\|u_n - u\| < \varepsilon, \qquad \forall n > N_\varepsilon$$

There is another, related notion of convergence:

Definition A.6 *Let $(V, \|\cdot\|)$ be a normed space. We say $\{u_n\}_{n\in\mathbb{N}} \subset V$ is a Cauchy sequence, if*

$$\lim_{n,m\to\infty} \|u_n - u_m\| = 0,$$

which is to say that, for every $\varepsilon > 0$, there is an integer N_ε, such that

$$\|u_n - u_m\| < \varepsilon, \qquad \forall n, m > N_\varepsilon.$$

It should be noted that the preceding two definitions are not equivalent. Certainly a convergent sequence must be a Cauchy sequence. (This is easily seen using the triangle inequality.) However, the converse implication is a bit more tricky. Depending on the choice of norm, a Cauchy sequence in V may converge to an element which is not in V. Here is an example:

Example A.6 Let $V = C([-1, 1])$, and consider functions $f_n : [-1, 1] \to \mathbb{R}$ defined by

$$f_n(x) = \begin{cases} 1 & -1 \le x \le 0 \\ 1 - nx & 0 < x < \frac{1}{n} \\ 0 & \frac{1}{n} \le x \le 1 \end{cases}$$

Let us define a norm by

$$\|f\| := \int_{-1}^{1} |f|\, dx.$$

(That this actually is a norm was shown in Example A.5.) One can easily show that

$$\|f_n - f_m\| \to 0, \qquad (n, m \to \infty)$$

Furthermore, if we define

$$f(x) = \begin{cases} 1 & -1 \le x \le 0 \\ 0 & 0 < x \le 1 \end{cases},$$

it is not difficult to show that

$$\|f_n - f\| \to 0, \qquad (n \to \infty).$$

But the function f is *not* continuous.

Is there a way to make the class of continuous functions a complete normed space? It turns out that the answer is affirmative:

Theorem A.1 *Let $X \subset \mathbb{R}^d$ be any closed and bounded set, and let $\|u\|_\infty = \max_{x\in X} |u(x)|$. The normed space $(C(X), \|\cdot\|_\infty)$ is complete.*

Proof Let $\{u_k\} \subset C(X)$ be a Cauchy sequence. Then, for any $\varepsilon > 0$, there is N_ε such that

$$\max_{x\in X} |u_k(x) - u_m(x)| < \varepsilon, \qquad \forall k, m > N_\varepsilon. \tag{A.19}$$

This implies that for any fixed $x \in X$, the sequence $\{u_m(x)\} \subset \mathbb{R}$ is a Cauchy sequence of real numbers. But the real numbers are complete. (Any finite dimensional space is.) Consequently

there is a limit in the real numbers, which we may call $u(x)$. Passing to the limit in (A.19), we have

$$\max_{x\in X} |u_k(x) - u(x)| \leq \varepsilon, \qquad \forall k > N_\varepsilon.$$

Now fix this value of ε, and some $k > N_\varepsilon$. Note that, by assumption, the function u_k is continuous. This means that we can find a $\delta > 0$, such that $|u_k(x) - u_k(y)| < \varepsilon$, for $|x - y| < \delta$. Now, for these x, y, consider

$$|u(x) - u(y)| \leq |(u(x) - u_k(x)| + |u_k(x) - u_k(y)| + |u_k(y) - u(y)| < 3\varepsilon.$$

Since ε is arbitrary, the function u is continuous. ■

Normed spaces in which *every* Cauchy sequence converges to a limit in the same space are important enough to have a special name reserved for them:

Definition A.7 *We say the normed space $(V, \|\cdot\|)$ is complete, if every Cauchy sequence in V converges to an element of V. A complete normed space is called a* Banach Space.

A.4.3 Linear Operators

Let V, W be linear spaces. A mapping $T : V \to W$ is linear if, for all $\alpha, \beta \in \mathbb{K}$ and all $v, w \in V$ there holds $T(\alpha v + \beta w) = \alpha T(v) + \beta T(w)$. (Note: For linear operators we often omit the brackets $(\cdot)$, i.e. we write Tu instead of $T(u)$.) If $W = \mathbb{K}$, we usually call T a *functional*.

Definition A.8 *Let V, W be normed spaces. We say a linear operator $T : V \to W$ is* bounded *(or* continuous*), if there exists a constant $C > 0$, such that for all $u \in V$,*

$$\|Tu\|_W \leq C \|u\|_V . \tag{A.20}$$

The reason why for linear operators T, boundedness is used synonymously with continuity is that linearity and boundedness imply

$$\|Tu - Tv\|_W = \|T(u - v)\|_W \leq C \|u - v\|_V .$$

In a normed space, we define continuity as the property that for each $\epsilon > 0$ there is a $\delta > 0$, such that $\|T(u) - T(v)\|_W < \epsilon$ for $\|u - v\|_V < \delta$. Obviously, for a bounded operator, we can set $\delta = \epsilon/C$ to satisfy this, where C is the constant in (A.20).

The smallest constant that satisfies (A.20) is the *operator norm* of T:

Definition A.9 *Let $T : V \to W$ be a continuous linear operator, then*

$$\|T\| := \sup_{\substack{u\in V \\ u\neq 0}} \frac{\|Tu\|_W}{\|u\|_V}$$

is bounded. We call this the operator norm *of T.*

Example A.7 Using standard definition of the matrix vector product in Euclidean vector space, a matrix can be viewed as a linear operator. Consider a square matrix of size n, then $A : \mathbb{R}^n \to \mathbb{R}^n$, $\mathbf{x} \mapsto A\mathbf{x}$ is a mapping which changes both the length and the direction of a vector. Let $\|\cdot\|_a$, $\|\cdot\|_b$ be any norm on $\mathbb{R}^n$. The operator norm

$$\|A\|_{a\to b} := \sup_{\substack{\mathbf{x}\in\mathbb{R}^n \\ \mathbf{x}\neq 0}} \frac{\|A\mathbf{x}\|_b}{\|\mathbf{x}\|_a} \tag{A.21}$$

measures the maximum change of length the matrix can cause when operating on any non-zero vector $\mathbf{x} \in \mathbb{R}^n$. (Often $\|\cdot\|_a$, $\|\cdot\|_b$ are the same.)

Definition A.10 *Let V be a Banach space. We say $V_0 \subset V$ is* dense *in V, if for every $v \in V$, there is a sequence $\{v_n\}_{n\in\mathbb{N}} \subset V_0$, such that*

$$\lim_{n\to\infty} \|v_n - v\|_V = 0$$

Example A.8 Let $I \subset \mathbb{R}$ be closed and bounded. The space $(C(I), \|\cdot\|_\infty)$, where $\|\cdot\|_\infty$ is the maximum norm, is a Banach space. Recall that by the Weierstrass approximation theorem, any continuous function $f \in C(I)$ can be approximated arbitrarily well by polynomials in maximum norm. The space $P(I)$ is dense in $C(I)$.

Theorem A.2 *Let V, W be Banach spaces, and let $V_0 \subset V$ be dense in V. Let $T : V_0 \to W$ be a continuous linear operator. The there exists a unique continuous linear operator $\widetilde{T} : V \to W$, such that $\widetilde{T}|_{V_0} = T$, and $\left\|\widetilde{T}\right\| = \|T\|$.*

A.4.4 Hilbert Spaces

An inner product is a mapping of two elements of a linear space to the field $\mathbb{K}$, which is endowed with certain properties:

Definition A.11 *Let V be a linear space. We call a mapping $(\cdot,\cdot) : V \times V \to \mathbb{K}$ an* inner product *(or: scalar product) if, for all $v, u, w \in V$, and $\alpha, \beta \in \mathbb{K}$, the following properties hold:*

$$\text{Linearity}: \quad (\alpha v + \beta w, u) = \alpha\,(v, u) + \beta\,(w, u) \tag{A.22}$$

$$\text{Conjugate Symmetry}: \quad (v, u) = \overline{(u, v)} \tag{A.23}$$

$$\text{Definiteness}: \quad (v, v) > 0 \text{ for } v \neq 0. \tag{A.24}$$

Note that if the linear space is defined of the field of real numbers, $\mathbb{K} = \mathbb{R}$, then conjugate symmetry reduces to ordinary symmetry, i.e. $(u, v) = (v, u)$.

Any linear space with a scalar product can be made a normed space, because we can define a norm in a canonical way using the inner product. Consider first the following simple example:

Example A.9 Let $V = \mathbb{R}^n$. Then

$$(\mathbf{x}, \mathbf{y}) := \sum_{i=1}^{n} x_i y_i$$

defines a scalar product. The standard Euclidean vector norm of Example A.5 (cf. eq. (A.16)) can be written in terms of the scalar product:

$$|\mathbf{x}|_2 = \sqrt{(\mathbf{x}, \mathbf{x})} = \sqrt{\sum_{i=1}^{n} x_i^2}.$$

It turns out that for any linear space V with an inner product $(\cdot, \cdot) : V \times V \to \mathbb{K}$, the mapping $V \ni x \mapsto \sqrt{(x, x)}$ defines a proper norm. In order to prove this, we need the following result (which is a very important result in its own right):

Theorem A.3 (Cauchy-Schwarz Inequality) *Let $(\cdot, \cdot)$ be a scalar product on a linear space V. Then, for all $u, v \in V$ there holds*

$$|\,(u, v)\,| \leq \sqrt{(u, u)}\sqrt{(v, v)}. \tag{A.25}$$

Proof For $v = 0$ there is nothing to prove. So assume $v \neq 0$, and consider $z \in V$ defined by

$$z := u - \frac{(u, v)}{(v, v)} v. \tag{A.26}$$

By construction, there holds $(z, v) = 0$. Consequently we have from (A.26):

$$\begin{aligned}
(u, u) = \left(z + \frac{(u, v)}{(v, v)} v, z + \frac{(u, v)}{(v, v)} v \right) &= (z, z) + \left| \frac{(u, v)}{(v, v)} \right|^2 (v, v) \\
&= (z, z) + \frac{|(u, v)|^2}{(v, v)} \\
&\geq \frac{|(u, v)|^2}{(v, v)}
\end{aligned}$$

This implies (A.25). ■

Now we are ready to prove that any norm induced by a scalar product is a proper norm:

Theorem A.4 *Let $(\cdot, \cdot)$ be a scalar product on the linear space V. Then the mapping $\|\cdot\| : V \to \mathbb{R}$,*

$$\|u\| := \sqrt{(u, u)} \tag{A.27}$$

defines a proper norm.

PROOF We need to show the properties (A.13) through (A.15). Definiteness follows from definition of a scalar product. Furthermore, by linearity of the scalar product, we have $\|\alpha u\| = \sqrt{|\alpha|^2 (u,u)} = |\alpha| \, \|u\|$ for all $\alpha \in \mathbb{K}$. Consider now for $u, v \in V$ arbitrary,

$$\begin{aligned}
\|u+v\|^2 = (u+v, u+v) &= \|u\|^2 + (u,v) + (v,u) + \|v\|^2 \\
&= \|u\|^2 + (u,v) + \overline{(u,v)} + \|v\|^2 \\
&= \|u\|^2 + 2Re\{(u,v)\} + \|v\|^2 \\
&\leq \|u\|^2 + 2|(u,v)| + \|v\|^2 \\
&\leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 \\
&= (\|u\| + \|v\|)^2 .
\end{aligned}$$

where we have used the Cauchy-Schwarz inequality. The triangle inequality (A.15) holds, and the proof is complete. ■

If a norm is defined as in Eq. (A.27), we say a norm is *induced* by the inner product.

Definition A.12 *A Banach space whose norm is induced by an inner product, is called a Hilbert space.*

The inner product allows us to define the concept of *orthogonality*:

Definition A.13 *Let V be a linear space, equipped with a scalar product. If $u, v \in V$ satisfy $(u,v) = 0$, we say u and v are orthogonal.*

Orthogonality is useful for example in association with bases for linear spaces:

Remark A.2 Let $v_1, \ldots, v_n$ be a basis for an n-dimensional linear space V. Then for any $v \in V$, we can write

$$v = \sum_{k=1}^{n} c_k v_k, \qquad c_k \in \mathbb{K}. \tag{A.28}$$

If the v_k are mutually orthogonal, i.e. $(v_k, v_j) = 0$ for $k \neq j$, and we furthermore assume $(v_j, v_j) = 1$ (which we can always achieve by appropriate scaling), then taking the scalar product with v_j, for some $1 \leq j \leq n$ on both sides of (A.28) leads to

$$\begin{aligned}
(v, v_j) &= \sum_{k=1}^{n} c_k (v_k, v_j) \\
&= c_j
\end{aligned}$$

An orthogonal basis thus makes it particularly easy to compute the coefficients in the representation (A.28).

Example A.10 A particularly simple example is the standard Cartesian basis for $\mathbb{R}^n$. (The Cartesian unit vectors are clearly orthonormal.) We can write any $\mathbf{x} \in \mathbb{R}^n$ as

$$\mathbf{x} = \sum_{i=1}^{n} x_i \mathbf{e}_i.$$

In this case the formula is clear by inspection, but it is worth noting that this conforms with our general representation formula, as $x_i = (\mathbf{x}, \mathbf{e}_i)$.